

The Exponential Class

- An important class of distributions, called the exponential class. This class possesses complete and sufficient statistics which are readily determined from the distribution.
- Regular exponential class: its pdf or pmf is

$$f(x; \theta) = \exp \left(p(\theta)K(x) + H(x) + q(\theta) \right), \quad x \in \mathcal{S}, \quad \theta \in \Omega(\text{an interval})$$

1. $\mathcal{S}$, the support of X , does not depend on θ
2. $p(\theta)$ is a nontrivial continuous function of θ
3. Continuous RV: $K'(x) \neq 0$ and $H(x)$ is a continuous function of x ; Discrete RV: $K(x)$ is a nontrivial function of x

Example 1: $N(0, \theta)$

Example 2: $Unif(0, \theta)$

- Let $X_1, \dots, X_n$ be i.i.d. from an exponential class, $Y_1 = \sum_{i=1}^n K(X_i)$ is a sufficient statistic for θ .
- Theorem: Let $X_1, \dots, X_n$ be i.i.d. from a regular exponential class, and $Y_1 = \sum_{i=1}^n K(X_i)$.
 1. The pdf or pmf of Y_1 is

$$f_{Y_1}(y_1; \theta) = R(y_1)e^{p(\theta)y_1 + nq(\theta)}$$

2. $E(Y_1) = -n \frac{q'(\theta)}{p'(\theta)}$
3. $Var(Y_1) = n \frac{1}{p'(\theta)^3} \{p''(\theta)q'(\theta) - q''(\theta)p'(\theta)\}$

Example 3: Poisson(θ)

- Theorem: Let $f(x; \theta)$, $\theta \in (\gamma, \delta)$ be a pdf or pmf of a RV X , whose distribution is a regular case of the exponential class. Then if $X_1, \dots, X_n$ is a random sample from the distribution of X , the statistic $Y_1 = \sum_{i=1}^n K(X_i)$ is a sufficient statistic for θ and the family $\{f_{Y_1}(y_1; \theta) : \theta \in (\gamma, \delta)\}$ of Y_1 is complete. That is, Y_1 is a complete and sufficient statistic of θ .
- If we can find $E(\varphi(Y_1)) = \theta$, then $\varphi(Y_1)$ is unique and is the MVUE of θ .

Example 4: $N(\theta, \sigma^2)$ with a known σ^2